NOTE

Paleomagnetism of the Hearst dike swarm and implications for the tectonic history of the Kapuskasing Structural Zone, northern Ontario

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Prominent north- to northwest-trending dikes (herein termed the Hearst swarm) west of the Kapuskasing Structural Zone (KSZ) in northern Ontario yield a paleomagnetic pole position of 48.6°N, 69.0°E, dm = 6.9, dp = 3.6 (60 samples from 12 dikes).

Using paleomagnetic, petrographic, and structural criteria the Hearst swarm is correlated with the 2.6 Ga Matachewan swarm east of the KSZ. A positive correlation is observed between the trend and paleomagnetic declination of Hearst–Matachewan dikes and may result from an ~15° post-2.6 Ga rotation of terranes across the KSZ.

Paleomagnetic and structural data from the Hearst dikes are consistent with the regional westward tilting of the crust near the southern KSZ as first proposed in 1983 by Percival and Card.

Des dykes de direction prédominante nord et nord-ouest (appelés ici le groupe de dykes de Hearst) à l'ouest de la zone structurale de Kapuskasing (ZSK) dans le nord de l'Ontario fournissent une position du pôle paléomagnétique de 48,6°N, 69,0°E, dm = 6,9, dp = 3,6 (60 échantillons prélevés sur 12 dykes).

En se servant des critères paléomagnétiques, pétrographiques et structuraux, nous corrélons le groupe de dykes de Hearst avec celui de Matachewan localisé à l'est de la ZSK et âgé de 2,6 Ga. Une corrélation positive est obtenue entre la direction et la déclinaison paléomagnétique des dykes de Hearst–Metachewan, ce qui semble être le résultat d'une rotation de ~15° des terrains au travers la ZSK à une époque plus récente que 2,6 Ga.

Les données paléomagnétiques et structurales des dykes de Hearst sont compatibles avec l'inclinaison régionale de la croûte vers l'ouest près de la partie sud de la ZSK, comme le proposaient en premier par Percival et Card en 1983.

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Introduction

In northern Ontario a north-northeast-trending zone of faulting cuts sharply across the east–west Archean grain and exposes deep crustal rocks at surface (Bennett *et al.* 1967; Thurston *et al.* 1977; Percival and Coe 1981). This feature, the Kapuskasing Structural Zone (KSZ), separates two Archean terranes with distinctly different geological and geophysical characteristics (Bennett *et al.* 1967). Diabase dikes of this region show a roughly orthogonal pattern (north to northwest and northeast). Their orientation, however, changes dramatically once the KSZ is crossed. The dike pattern on the western side is rotated up to ~40° counterclockwise compared with that on the eastern side (Fig. 1). An accompanying change in the trend of the Archean structural grain also occurs and suggests that relative rotation between two crustal blocks has occurred as a result of fault movements associated with the KSZ (Bennett *et al.* 1967). Such a rotation, however, is not apparent in the extreme northern part of the KSZ where dike aeromagnetic anomalies appear to cross the KSZ with little change in trend.

The regional metamorphic grade increases eastwards from greenschist grade in the Michipicoten greenstone belt (near Wawa) to granulite grade within the KSZ northeast of Chapleau. This observation, amongst others, suggests that upthrusting along the eastern margin of the southern KSZ has resulted in a regional westward tilting exposing an oblique section through 20 km of Archean crust (Percival and Card 1983).

Dikes on the eastern side of the KSZ comprise: (1) 2.6 Ga north-trending Matachewan quartz tholeiites, (2) 2.15 Ga northeast-trending Preissac tholeiites, and (3) 1.15 Ga east-northeast-trending Abitibi alkali olivine basalts (Fahrig 1972; Card *et al.* 1981; Gates and Hurley 1973; Hanes and York 1979). Although several paleomagnetic studies have been done on these dikes (e.g., Strangway 1961, 1964; Larochelle 1966; Irving and Naldrett 1977; Buchan and Schwarz 1981; Schutts and Dunlop 1981), a clear picture of variation in primary direction and polarity within any one swarm has yet to emerge.

Comparatively little is known about the dikes west of the KSZ. Information is largely confined to reconnaissance petrography and petrochemistry (Siragusa 1977; Thurston *et al.* 1977; Card *et al.* 1981). K–Ar ages on NW-trending dikes range from 2.4 to 1.4 Ga (Wanless *et al.* 1966; W. F. Fahrig, personal communication, 1984). On the basis of aeromagnetic data Watson (1980) correlated the northwest- and northeast-trending dikes with the Matachewan and Preissac (or Abitibi), respectively, whereas Thurston *et al.* (1977) considered the northwest-trending dikes to be part of the 1.25 Ga Mackenzie swarm (located to the northwest mainly in the northwest Superior, Churchill, Slave, and Bear provinces, Fahrig and Jones 1969).

This paper focuses on the north- to northwest-trending dikes west of the KSZ, assessing their (1) paleomagnetic characteristics, (2) relationship to the Matachewan swarm, and (3) implications for the tectonic history of the KSZ.

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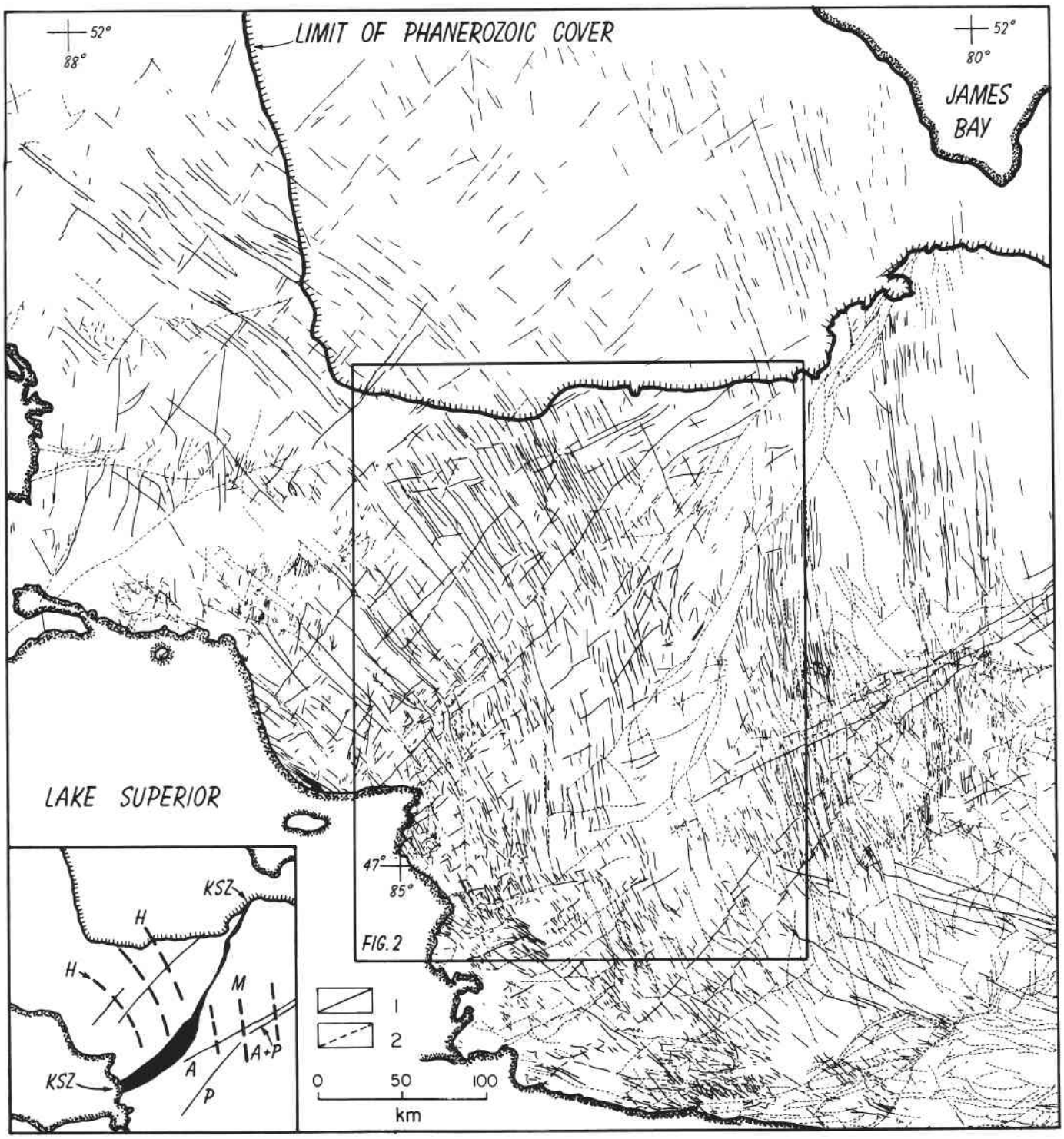


FIG. 1. Map showing the distribution of diabase dikes in northeastern Ontario (after Ontario Division of Mines maps 2199 (1971) and 2393 (1978). Legend: 1 = diabase dike; 2 = fault. The various dike trends are labelled in the inset map as follows: M = Matachewan; A = Abitibi; P = Preissac; H = Hearst. The location of the study area (Fig. 2) is also shown.

Method

Samples for petrographic and paleomagnetic study were obtained from 22 dikes (Fig. 2). Site distribution was designed to test any spatial variation in structural, petrographic, and magnetic properties with respect to the KSZ.

Block samples were oriented in the field using both magnetic and sun compasses and cored in the laboratory prior to measurement on a shielded DIGICO spinner magnetometer. Cores were subjected to alternating field (AF) demagnetization to 100 mT and thermal treatment to 580°C using Schonstedt demagnetizers.

Magnetic components were resolved using a combination of vector diagrams (Zijderveld 1967), remagnetization circles (Halls 1976), and principal component analysis (Kirschvink 1980). Further details are given in Ernst (1981).

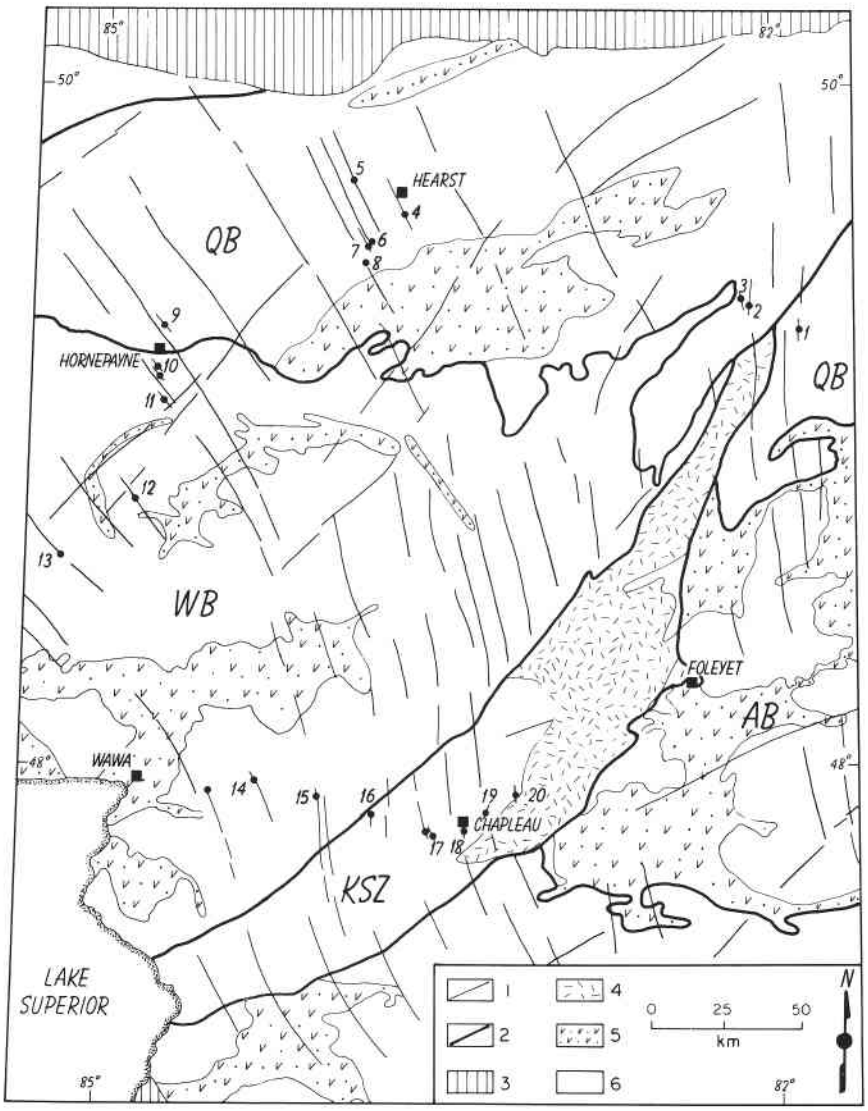


FIG. 2. Geological map (after McCrank *et al.* 1980) showing the location of paleomagnetic sampling sites. Sites without a number gave inconsistent data and were rejected. Legend: 1 = dike; 2 = structural subprovince boundary; 3 = post-Archean cover rocks; 4 = granulite to amphibolite grade gneiss, gabbro, and anorthosite; 5 = supracrustal volcanic and sedimentary rocks; 6 = granite or granite gneiss. Abbreviated names on map: QB = Quetico Metasedimentary Belt; WB = Wawa Volcanic Belt; AB = Abitibi Volcanic Belt; KSZ = Kapuskasing Structural Zone.

Petrography and structure

The dikes of this study have near-vertical dips and thicknesses that range from less than 0.05 m to more than 70 m; the average thickness is about 30 m. Lengths of individual aeromagnetic dike anomalies reach 200–300 km. The calculated crustal extension indicated by this swarm is 1.0%, based on an average dike thickness of about 30 m.

Dike interiors are fine to medium grained with intergranular to subophitic textures that consist of subequal amounts (about 40% each) of weakly zoned saussuritized plagioclase (labradorite) and clinopyroxene (mostly augite with minor pigeonite). Pyroxenes are variably altered to chlorite and uralite and many are also rimmed by blue–green amphibole. Interstitial quartz is always present and composes up to 7% of the rock, but olivine is absent. Minor biotite and apatite also occur. Opaque phases (up to about 10%) include Ti magnetite, ilmenite, and trace sulphides. The Ti magnetite usually shows ilmenite exsolution lamellae and may contain fine hematite rods, suggesting a low-level deuteric oxidation state of C3 (Haggerty 1976). Phenocrysts of greenish plagioclase up to 3 cm diameter are common but their distribution is very heterogeneous when viewed on the scale of outcrop, dike, or swarm. They are frequently concentrated near dike margins or apophyses (e.g., dike 11), but some dikes (e.g., 8 and 15) have a median phenocryst-rich layer that may be a final phase of injection. Dike 4 contains alternate plagioclase-free and -poor layers, about 1 cm wide, that parallel the dike margins.

The north- to northwest-trending dikes are classified as plagioclase – porphyritic quartz tholeiites on the basis of modal mineralogy. Their textural and mineralogical similarities suggest a single intrusion event, identified in this paper as the Hearst swarm.

Paleomagnetic results

On AF demagnetization Hearst dikes decay rapidly in intensity, having median destructive fields of about 8 mT. Above 30–50 mT a characteristic component for the dike suite is revealed as a stable end point (SEP) corresponding to a linear

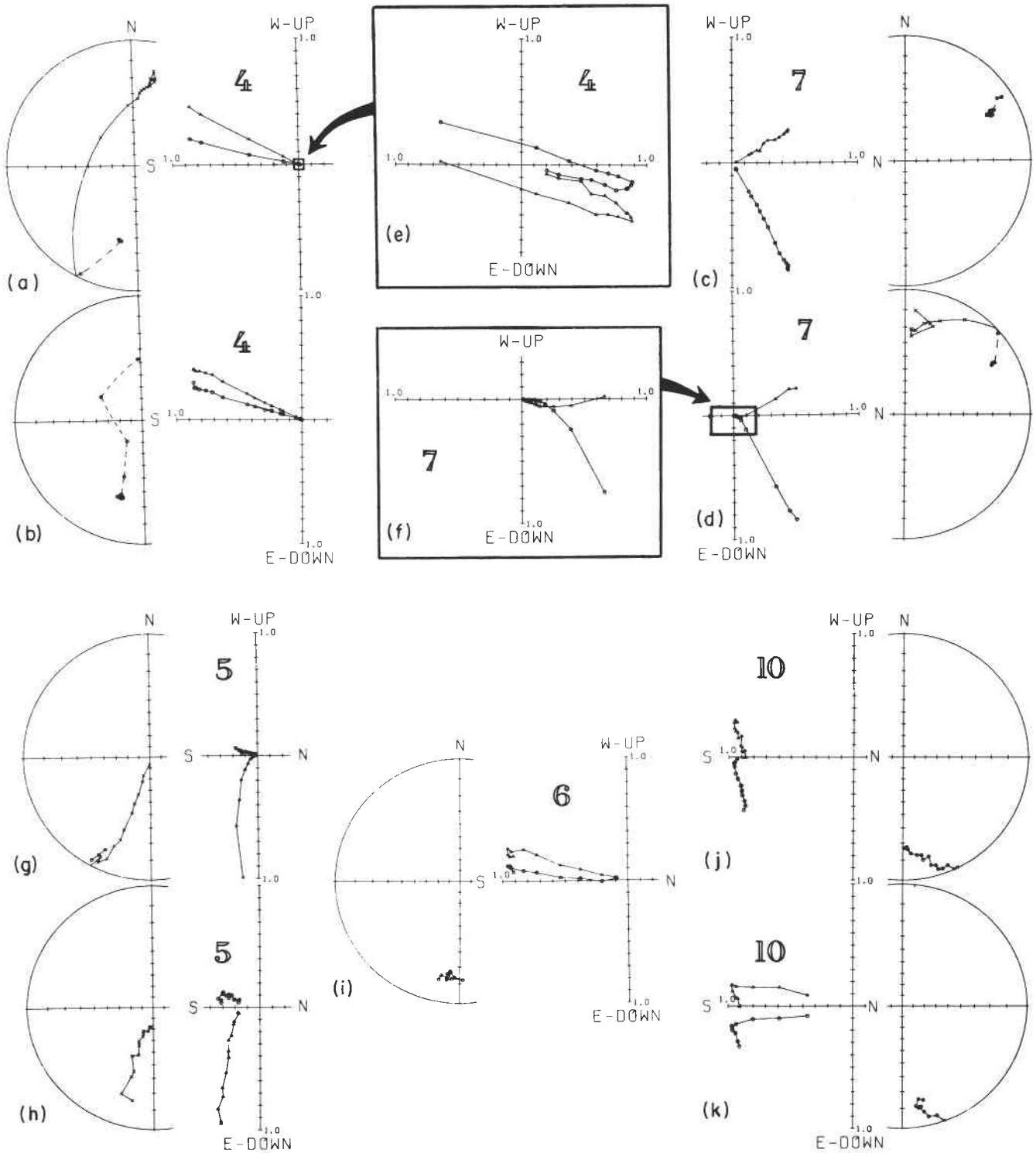


FIG. 3. Examples of stereoplots and vector diagrams showing behaviour of dike specimens under AF and thermal demagnetization. On stereoplots open circles (crosses) are upward- (downward)-pointing magnetizations; on vector diagrams circles (triangles) represent projections of magnetization vector onto the horizontal (vertical) plane. All demagnetization data shown are either taken to the limit of AF demagnetization (100 mT) or terminated once erratic behaviour begins. For all thermal data that are not shown decaying to the origin a 5°C increment above the maximum temperature gives results in an immediate decay to the origin and a spurious magnetization direction. (*a*) Hearst dike margin, site 4, AF to 35 mT, each small-scale division (SD) on vector diagram = 214×10^{-3} A/m (all SD intensities below expressed $\times 10^{-3}$ A/m). (*b*) Same sample, alternate specimen, thermal to 550°C, SD = 282. (*c*) Hearst dike interior, site 7, thermal to 560°C, SD = 2294. (*d*) Same sample, alternate specimen, AF to 100 mT, SD = 2334. (*e*) Enlargement of low-intensity area of vector diagram in (*a*), AF 14 → 98 mT, SD = 15. (*f*) Similar enlargement for vector diagram in D, AF 8–100 mT, SD = 350. (*g*) Hearst dike margin, site 5, AF to 35 mT, SD = 282. (*h*) Same sample, alternate specimen, thermal to 550°C, SD = 214. (*i*) Hearst dike interior, site 6, AF to 100 mT, SD = 1.8. (*j*) Hearst dike margin, site 10, thermal to 560°C, SD = 116. (*k*) Same sample, alternate specimen, AF to 25 mT, SD = 107.

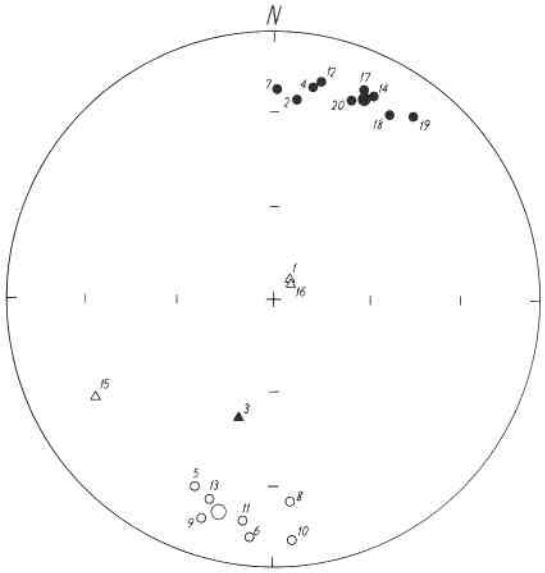


FIG. 4. Equal-area stereonet showing paleomagnetic site directions (Table 1) for Hearst (●, ○) and anomalous dikes associated with Kapuskasing Structural Zone (▲, △). Mean site directions (●, ○) are shown for the two polarity groups of the Hearst swarm. Open (closed) symbols represent upward (downward) magnetizations.

segment passing through the origin of vector diagrams (Fig. 3).

This component is often not isolated until more than 90% of the natural remanent magnetization (NRM) has been removed. During demagnetization the remanence vector commonly follows a great-circle path on the stereonet (e.g., Fig. 3*a*), indicating the presence of an additional component. Sample remagnetization circles have no consistent orientation within a site, signifying that the soft component (probably a viscous remanent magnetization (VRM)) is widely dispersed and of no paleomagnetic significance. In some samples remagnetization circles are generated without attainment of an SEP. Mean site directions are computed using both SEP and remagnetization circle data according to the method of Bailey and Halls (1984). Thermal demagnetization proved less capable than AF of isolating the characteristic component because overprints sometimes survive the highest unblocking temperatures (see Fig. 3).

The characteristic component is considered to be primary because it is typically isolated at the highest coercivities (50−100 mT) and unblocking temperatures (550−570°C) and is best preserved in fine-grained and chilled margin samples having the least silicate alteration.

In order to further test the primary nature of the remanence, baked and unbaked country rock samples were also collected at three dike sites (5, 11, and 12) but the results were inconclusive (Ernst 1981). Apart from four anomalous dikes, all in the vicinity of the KSZ, the directions of the characteristic component fall into two approximately antiparallel populations (Fig. 4, Table 1). For any one dike the polarity of the component is the same for all samples, suggesting that intrusion of the Hearst swarm spanned a reversal of the Earth's magnetic field. In some samples there is a soft, intense component, opposite to the characteristic one (e.g., Fig. 3 *a* and *e*) that may be due to self-reversal. Although there are about equal numbers of normal and reversed polarity dikes, there is some suggestion of a higher proportion of normal ones closer to the KSZ.

An important aspect of the two populations of Hearst directions shown in Fig. 4 is that even allowing for stereonet distortion they are both streaked in declination. This smearing of directions is unlikely to reflect unremoved secondary components because demagnetization paths in general do not follow flat, low-inclination trajectories. Instead, there is evidence, discussed below, that the streaking is due to deformation associated with the evolution of the KSZ.

Discussion

Correlation of Hearst and Matachewan swarms

The Hearst and Matachewan dike swarms are paleomagnetically very similar: they yield matching mean site directions (Table 1) and have a comparable disposition to normal and reversed polarities (Ernst 1981). Both swarms are quartz tholeiites with characteristic saussuritized plagioclase phenocrysts and have similar mineralogy and alteration. On average, dike thickness and spacing are also comparable for the two swarms (Ernst 1981).

These paleomagnetic, petrologic, and structural similarities indicate that the Hearst dikes represent a northwest continuation of the Matachewan swarm as previously proposed by Watson (1980). A series of almost unbroken linear aeromagnetic anomalies extends northwestwards from the area sampled to near the Berens−Uchi subprovince boundary west of James Bay (see ODM−GSC 1970). Some of these anomalies can be correlated with individual dikes of this study. On this basis, the Hearst−Matachewan swarm has a longitudinal extent exceeding 700 km and covers an area greater than 2×10^5 km^2.

Tectonic implications

There is a positive correlation between dike trend and the declination of the characteristic component. Northwest-trending Hearst dikes have a mean remanence direction rotated ~15° counterclockwise from both north-trending Hearst dikes near the southern KSZ and the north-trending Matachewan dikes (Table 1). The directions are statistically different at the 0.05 confidence level (statistics from Tarling 1983). One possible explanation is that the northwest dikes have been rotated counterclockwise relative to the north dikes during deformation along the KSZ, as suggested by Bennett *et al.* (1967). However, the declination difference (~15°) is less than the change in dike trend (25−40°), so some of the variation in dike trend must be primary. Other explanations for the paleomagnetic data are that the northwest-trending dikes are of a slightly different age, or that the remanence vector is biassed into the plane of the dike (Strangway 1961). This latter possibility needs to be checked by detailed rock magnetic studies but was not revealed in a statistical study of worldwide paleomagnetic data obtained from dikes (Evans 1968).

Structural and paleomagnetic data from the Hearst dikes support the model of crustal tilting associated with KSZ thrusting proposed by Percival and Card (1983) and indicate that this deformation occurred at least in part subsequent to dike intrusion (i.e., after 2.6 Ga). The region affected by tilting is marked by a dike-dip reversal from steeply southwest to east as the KSZ is approached (Table 1, Fig. 5*b*) (Ernst 1981, 1982). Sites 17, 18, 19, and 20 are all within this region of easterly dips. The paleomagnetic direction of these north-trending Hearst dikes differs by 12° from the mean direction of the Matachewan swarm as determined by Schutts and Dunlop (1981) (directions are statistically different at the 0.05 level) and by 6° in the same sense from the mean Matachewan direc-

TABLE 1. Summary of paleomagnetic data

Site	θ (°)	ϕ	*d* (m)	P	N_T	N_S	N_R	*D* (°)	*I* (°)	α_{95} (°)	*k*
					Hearst swarm						
2[a]	345	S	15	N	3		3	6.2	27.3	—	—
4[a]	335	S	>30	N	3		3	10.6	22.6	—	—
5	335	85°E	8	R	6/1		6	202.5	−25.4	12.6	110
6[a]	350	S	50	R	5	2		185.6	−12.6	—	—
7	315	S	30	N	5		5	0.8	24.0	4.5	1300
8[a]	335	S	55	R	9	6	2	175.0	−25.6	24.9	30
9	320	80°W	34	R	6		5	197.7	−15.8	15.5	120
10	330	87°W	0.6	R	4	3		174.8	−10.8	4.6	>20
11	320	80°W	28	R	6/2	6	2	187.5	−18.4	4.8	160
12	330	80°W	29	N	7/3	3	5	12.2	19.9	5.0	220
13	315	90°	46	R	4	1	3	197.7	−22.9	12.4	190
14	325	85°E	24	N	6/1	2	3	26.1	18.6	11.5	90
17	345	76°E	3	N	5	5		23.7	18.4	5.3	210
18	05	82°E	9	N	3	3		32.3	20.0	7.0	310
19	15	80°E	>20	N	5	5		36.9	15.4	4.6	280
20	340	75°E	>3	N	4	3		21.7	23.0	15.1	70
Mean of N polarity dikes[b]					7			23.4	21.3	8.6	50
Mean of R polarity dikes[b]					5			193.9	−18.5	11.6	40
Mean of NW-trending dikes[b,c]					7			12.3	19.4	8.2	60
Mean of N-trending dikes[b,d]					4			29.7	20.8	8.5	120
Mean direction[b]					12			19.4	20.2	6.6	40
					Pole position: P_{LAT} = 48.6°N, P_{LONG} = 69.0°E dm = 6.9°, dp = 3.6°						
					Anomalous Hearst dikes						
1	0	S	15	R	2	2		44.5	−82.8	—	—
3	335	70°W	14	N	6	3		196.0	51.7	4.2	860
15	345	77°W	24	R	7		7	241.0	−25.3	13.4	60
16	355	85°E	31	R	7	5		49.6	−84.2	12.2	40
					Matachewan swarm						
Irving and Naldrett (1977)[e]					10			27	16	7	50
Schutts and Dunlop (1981)[e]					14			21	13	6	40

NOTES: θ, ϕ, and *d* are, respectively, dike trend, dip, and thickness; S signifies that the dike dip is probably steep (ϕ > 80°) but no measurement could be made; P = magnetic polarity with respect to present field; N_T = number of dike samples measured/number of country rock samples measured and used in mean *or* number of paleomagnetic sites; N_S = number of samples yielding linear segments on vector diagrams (all corresponding to stable end points); N_R = number of other samples yielding remagnetization circles; *D*, *I* = declination and inclination at collecting site; α_{95} = radius of 95% confidence circle or radius of circle with area equivalent to that of the 95% confidence ellipse for sites including remagnetization circle data analysed by the method of Bailey and Halls (1984); *k* is Fisher's precision parameter (for sites including remagnetization circle data, *k* is calculated using the method of Bailey and Halls 1984); dm, dp are semi-axes of 95% confidence oval about the pole position.

[a]These sites have not been used in the mean site calculations. Sites 2 and 4 are omitted because each is based on only three remagnetization circles and no stable end points, site 8 is omitted because of its large α_{95}, and site 6 is omitted because it is based on only two samples.

[b]Mean declination and inclination obtained from *D*, *I* values all recalculated for the location of Timmins, Ontario (lat. 48.5°N, long. 81.3°W).

[c]All are northwest-trending dikes distant from KSZ and associated deformation and include sites 5, 7, 9, 10, 11, 12, and 13.

[d]All are north-trending dikes within region affected by KSZ block tilting and include sites 17, 18, 19, and 20.

[e]These results come from sites near Timmins and thus can be directly compared with the mean direction for the Hearst swarm (see footnote *b*).

tion of Irving and Naldrett (1977) (Table 1). Assuming that these north-trending Hearst and Matachewan means were originally coincident, the present difference can be modelled by an ~15° westward tilting of the Hearst data about a horizontal axis, approximately parallel to the KSZ boundary faults. This rotation is similar both in direction and magnitude to that advocated by Percival and Card (1983).

Hearst dikes tend to be thinner and less numerous in the vicinity of the KSZ and are conspicuously absent from the Shawmere anorthosite and other granulite facies rocks of the KSZ (Figs. 2, 5) (Ernst and Halls 1983; Thurston *et al.* 1977). A possible cause is that the dikes taper downwards and finally pinch out entirely at crustal depths corresponding to granulite grade conditions.

Conclusions

A reconnaissance paleomagnetic and petrographic study of north- to northwest-trending dikes west of the Kapuskasing Structural Zone suggests their correlation with the 2.6 Ga old north-trending Matachewan swarm to the east. The Hearst–Matachewan swarm is thus more than 700 km long and is characterized by a gradual change from north- to northwest-trending dikes as it is followed northwards across the Kapuskasing Zone. This change in trend may in part be caused

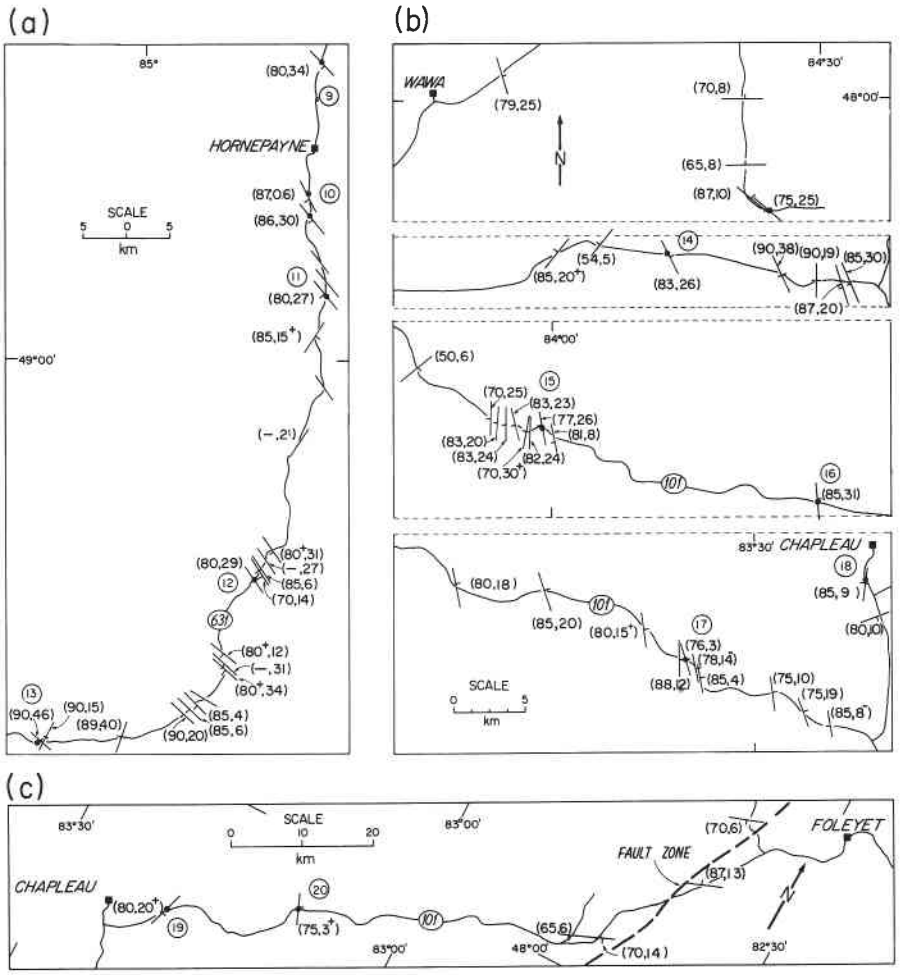


FIG. 5. Maps of structural data from dikes exposed along (*a*) Hwy 631 through Hornepayne; (*b*) Hwy 101 between Wawa and Chapleau; and (*c*) Hwy 101 between Chapleau and Foleyet. Figure 2 shows locations. Numbers in parenthesis refer to the dip (degrees) and thickness (metres) of each dike (+, − after thickness means greater than, less than). Black dots are paleomagnetic sites; those not marked by a circled number were rejected. The uniformity of dike attitude in (*a*), remote from the KSZ, is to be compared with (*b*), where there is a change from predominantly west- to east-dipping dikes on approaching the KSZ from the west. Note that dikes in the vicinity of the KSZ tend to be relatively thin, and that only east-northeast-trending dikes of a probably younger age occur in the granulite terrane (*c*).

by subsequent deformation or may be an entirely original feature reflecting a slightly different time of emplacement between the northwest- and north-trending swarm segments.

When combined with structural observations the paleomagnetic data suggest that a post-2.6 Ga westward tilting of the crust has taken place in the vicinity of the Kapuskasing Zone, consistent with the tectonic model proposed by Percival and Card (1983).

Acknowledgments

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